



Traffic crash data augmentation with multi-type variables using hybrid VAE-Diffusion generative neural networks for enhancing crash frequency modeling

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ABSTRACT

Crash frequency modeling aims to analyze influential factors of crashes to enhance road safety. However, as crashes are inherently rare events, excessive zero observations in crash datasets undermine crash frequency models' ability to identify high-risk road segments. Existing statistical models are often limited by strict distributional assumptions, while resampling and deep generative methods often distort data representation or struggle with multi-type crash data (count, ordinal, nominal, and real-valued) since these heterogeneous variables follow different statistical distributions and require distinct encoding strategies. This study proposes TabSyn, a hybrid VAE-Diffusion model that transforms multi-type crash data into a continuous latent space through a tokenizer-based embedding and VAE encoding, enabling the model to preserve the inter-variable correlations and underlying data structure. To validate its effectiveness, TabSyn is compared with state-of-the-art generative models (CTGAN, TVAE, GReaT, and StaSy) in synthetic data quality, and integrated with XGBoost for crash frequency prediction against two statistical models (ZIP, GAM-Poisson). Results demonstrate that TabSyn outperforms benchmark methods by achieving the best synthetic data distributional and structural fidelity, and the lowest prediction error, particularly for Top-K high-risk segment identification. This study offers valuable insights for improving crash frequency modeling and traffic safety management through imbalanced data augmentation of multi-type crash data.

1. Introduction

Crashes are complex and multidimensional events where many interrelated factors can contribute to their occurrence. To account for the complexity and randomness of crash data, crash frequency modeling is widely employed as a useful tool to analyze how various explanatory factors, e.g., traffic volume, road geometry, and environmental conditions, correlate to the aggregated number of crashes that occurred in a given time at different locations (Lord et al., 2021). Accurate crash frequency modeling is essential in assisting governmental agencies to develop effective safety policies and allocate resources to high-risk areas, ultimately improving traffic safety and reducing crashes (Lord and Mannering, 2010). However, the crash data are often characterized by excess zero observations, indicating that a significant portion of the data consists of zero crash counts (Ali et al., 2024; Dzinyela et al., 2024). This can be problematic in crash frequency modeling because

excess zeros can bias parameter estimates, resulting in erroneous predictions and misleading inferences about explanatory factors (Raihan et al., 2019).

The root of having excess zero observations in crash data lies in the limitations of the traditional data collection method. Crash data are often collected from police reports, insurance claims, and hospital records after crashes occurred and the data collection process is typically costly (Lord et al., 2021). Given that crashes are low-probability events, post-crash data collection methods naturally lead to a small sample size of recorded crashes. Moreover, many minor crashes are not documented due to factors such as perceived insignificance or absence of police involvement, which further reduces the sample size of recorded crashes (Lord et al., 2005). These limitations collectively result in a crash count distribution that skews heavily toward zero. Although there are emerging technologies, such as data

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collection from on-road electronic equipment like video surveillance, radars, loop detectors (MahaVishnu et al., 2017), and from onboard equipment installed on vehicles to gather crash data related to vehicle conditions, driver distractions, traffic information, etc (Josephinshermila et al., 2023; Han and Abdel-Aty, 2025). However, since crashes are rare, such collection methods still struggle with the problem of excess zero observations, as they cannot capture data where crashes are nonexistent.

Approaches to address excessive zero observations in crash data include statistical models, algorithm-level machine learning (ML) methods, and data-centric techniques. Statistical models, such as zero-inflated Poisson and zero-inflated negative binomial (Lee and Mannerling, 2002) are often constrained by assumptions of data distribution and may not fully capture the underlying patterns in crash datasets with multiple data types (i.e., count, ordinal, nominal, and real-valued) and complex interactions between various explanatory variables (Ding et al., 2022). Algorithm-level machine learning (ML) methods, such as cost-sensitive learning and ensemble ML models tailored for class imbalance (e.g., AdaBoost), are criticized for the subjectivity in tuning appropriate costs (Wang et al., 2016) and for being inferior to data-centric resampling techniques, often struggling with limited generalization ability (Man et al., 2022b; Galar et al., 2012). Data-centric techniques provide greater flexibility and effectiveness in handling excessive zero observations. Currently, the predominant data-centric strategies are resampling techniques, including under-sampling and over-sampling (Pei et al., 2016). However, under-sampling can discard potentially valuable data, while over-sampling methods, such as SMOTE and its variants, may generate synthetic minority samples that inadequately capture the minority-class distribution and lie close to majority-class samples, which can blur class boundaries and degrade model performance in real-world applications (Tarawneh et al., 2022).

Deep generative models are effective techniques in alleviating excessive zero observations; meanwhile, they can reduce the time-consuming process of collecting large-scale real data by generating synthetic records, especially because crash data is naturally scarce and limited. Deep generative models show strong performance on image data with continuous pixel values and local spatial correlations, yet face challenges with heterogeneous tabular data due to its multi-type features and lack of inherent spatial relationships (Borisov et al., 2024). Several deep generative models show improvement in tabular data generation, such as Conditional Tabular Generative Adversarial Networks (CTGAN) (Xu et al., 2019), tabular variational autoencoder (TVAE) (Xu et al., 2019), and Generation of Realistic Tabular data (GReaT) (Stoian et al., 2024). Recently, a novel Diffusion model has demonstrated even greater performance in generating data with high quality and diversity (Kim et al., 2023). Despite its impressive capabilities, applying diffusion models to tabular data still presents challenges (Kotelnikov et al., 2022). One of the key challenges is that encoding techniques commonly used in deep generative models face difficulties in handling tabular crash data (Borisov et al., 2024). Most existing models adopted simple encoding techniques such as one-hot encoding (Lee et al., 2023; Liu et al., 2023) and analog bit encoding (Zheng and Charoenphakdee, 2022) to convert discrete features into numerical format. However, the crash dataset is highly structured multi-type data with high-cardinality variables such as count and ordinal variables, which simple encoding methods struggle to handle. Simple encoding may aggravate the “curse of dimensionality” problem, expanding tabular crash data into high-dimensional vectors. This dimensional expansion in the data increases computational complexity as the models struggle to capture meaningful correlations across features. Consequently, this undermines the model’s ability to generate realistic synthetic data (Lee et al., 2023; Liu et al., 2023).

To this end, this study introduces a data-driven crash frequency modeling approach based on TabSyn, a hybrid VAE-Diffusion neural network (Zhang et al., 2024). Following data generation, we compare the proposed model with the baseline generative models, in terms of the

synthetic data quality and prediction accuracy. Lastly, we interpret the prediction result and provide policy recommendations for traffic safety and management. The main contributions of the study are summarized below:

(1) A novel deep generative neural network is proposed to process multi-type crash data by transforming diverse input features into a unified embedding space, enabling the generation of synthetic data that accurately reflects real-world crash distributions and effectively addresses excessive zero observations.

(2) A comprehensive study was conducted to evaluate the quality of synthetic data using both coverage and structural similarity metrics, and to compare the performance in crash frequency modeling of various models (e.g., CTGAN, TVAE, GReaT, StaSy, ZIP, GAM-Poisson) to the proposed model. Results show that the proposed model outperforms all baseline models.

2. Literature review

2.1. Challenges of crash frequency modeling

Crash frequency modeling serves as a vital tool for analyzing non-negative and discrete crash events on roadways, enabling the assessment of safety performance across entities such as roadway segments, intersections, and corridors (Lord et al., 2021). Conventional statistical models for handling excessive zero observations often adapt count-data models to better accommodate the large number of zero observations. Raihan et al. (2019) and Lord et al. (2005) applied zero inflation Poisson model and the zero-inflation negative binomial model, which assume two processes: one where the roadway section is virtually safe with zero crashes, and another where the section is in a non-negative count state for crashes, following a Poisson or negative binomial distribution. Yet, according to Lord et al. (2007), these models have intrinsic logical shortcomings, as they fail to establish clear boundary conditions to differentiate fully safe states from unsafe states. Subsequently, the negative binomial Lindley model, a mixture of negative binomial and Lindley distributions (Ghitany et al., 2008; Lindley, 1958), was introduced to address excess zero observations (Behara et al., 2021; Rusli et al., 2018; Shinthia Azmeri et al., 2023). Building on this, Islam et al. (2022) employ a Finite Mixture Negative Binomial-Lindley (FMNB-L) model, combining the Negative Binomial-Lindley distribution with finite mixture modeling to account for heterogeneous subpopulations and excess zero observations in crash data. However, similar to other statistical models, a notable limitation of this approach is that the increased complexity of the model restricts its applicability to relatively small datasets. Owing to the limitations of statistical models, such as predefined assumptions and inability to handle large datasets, machine learning-based approaches have gained traction for crash frequency modeling.

Compared to statistical models, machine learning models are flexible, do not require prior assumptions about data distribution, and can learn complex patterns in high-dimensional data (Zeng et al., 2016). For example, CART (Classification and Regression Trees), a decision tree-based approach that recursively splits data into subsets was employed to model crash frequency. The method achieved better prediction accuracy compared to a negative binomial regression model, with accuracy rates of 52.6% and 52.3%, respectively (Chang and Chen, 2005). Das and Abdel-Aty (2011) model crash frequency using a Genetic Programming (GP) approach and conclude that GP is effective in capturing the relationship between crash frequency and explanatory factors such as average daily traffic, roadway conditions, and skid resistance, providing a comprehensive understanding of rear-end crashes on urban arterials. An optimized Neural Network (NN) approach was employed to model crash frequency and not only outperforms traditional Negative Binomial (NB) models in terms of fitting and predictive performance but also effectively identifies and removes variables with insignificant

effects on crash frequency, resulting in a more interpretable and simplified model (Zeng et al., 2016). Additionally, the extracted rules confirm the existence of nonlinear relationships between crash frequency and explanatory factors. Although machine learning models excel at capturing complex nonlinear relationships in crash frequency modeling, their performance is still limited by data quality, including the completeness and representativeness of the available data (Mannering et al., 2020; Han et al., 2024).

2.2. Challenges of resampling techniques for multi-type crash frequency data

Previous studies focusing on data-centric approaches often utilize data aggregation or resampling techniques to solve excessive zero observations. Data aggregation involves adjusting space and time scales to reduce the number of zero observations while maintaining the homogeneity of the observations (e.g., lane width, shoulder width, etc.). For example, aggregate 200-meter road segments into 400-meter segments, combined with aggregation over different time scales (e.g. 2, 3, 4, and 6 years) to decrease the proportion of zero observations before developing regression models. However, this may lead to a loss of spatial-temporal information and compromise the accuracy of crash frequency prediction (Usman et al., 2011). Traditional resampling techniques are more sophisticated methods to address excessive zero observations. Abdel-Aty et al. (2004) and Yang et al. (2018) use matched case-control design to select a matched sub-set out of the full dataset, controlling for external factors such as time of day, season, year, geometric and roadway features, etc. Yet such under-sampling methods may lead to a loss of important information and potentially degrade model performance (Cai et al., 2020). On the other hand, Chen et al. (2022) applied the Synthetic Minority Over-sampling Technique for panel data (SMOTE-P), which increases minority class samples to balance the data, to deal with excessive zero observations of bus-involved crashes and improve model accuracy. Nevertheless, since SMOTE uses interpolation of existing samples, it can oversimplify relationships between variables, generating synthetic data that fails to capture the full complexity of crash patterns or reflect real-world variability, potentially leading to overfitting or biased predictions (Branco et al., 2016; Tarawneh et al., 2022). This limitation has motivated the adoption of deep generative models that can better represent and address the complexities inherent in crash frequency data (Chen et al., 2024).

Crash frequency data often contain multi-type variables (i.e., count, ordinal, nominal, and real-valued variables), posing significant challenges for deep generative models in addressing excessive zero observations. Most existing studies use Generative Adversarial Networks (GANs) (Cai et al., 2020; Man et al., 2022a) or variational autoencoder (VAE) (Ding et al., 2022) to address class imbalance in crash data. As deep generative models evolve, various models built on different foundational architectures emerge (Zhang et al., 2024). Although these methods offer advantages for multi-type data synthesis, concerns remain about data quality due to their limitations in capturing the complex relationships between variables. Conditional Tabular Generative Adversarial Networks (CTGAN) and tabular variational autoencoder (TVAE) (Xu et al., 2019) both utilize the Mode-specific Normalization model and treat each category of discrete variables as a separate mode and normalize the continuous variables within each mode. However, this method can introduce model complexity that can obscure the features of the variables and may fail to capture potential relationships between different categories. Score-based Tabular Data Synthesis (StaSy) (Kim et al., 2023) is a diffusion-based model, which employs a technique that treats one-hot encoded categorical variables as continuous features and processes them together with numerical columns. Nevertheless, treating all variables as continuous may result in high-dimensional feature spaces and make it difficult for the model to learn data features and capture intrinsic features of the discrete variables.

Generation of Realistic Tabular data (GReaT) (Stoian et al., 2024) employs a serialization technique that converts both discrete and continuous variables into natural language sentences. Then the ‘sentences’ can be processed by an auto-regressive GPT model. However, the natural language descriptions can lead to loss of detail or ambiguity, as similar values might be described in similar ways, affecting the model’s ability to capture precise data features. Additionally, while shuffling rows ensures permutation invariance, it may not fully capture the relationships and structures between columns, particularly when column order is important for analysis.

3. Methodology

3.1. Multi-type data generative neural network

We develop our crash data augmentation approach based on TabSyn, a deep generative model for multi-type tabular data synthesis (Zhang et al., 2024). As shown in Fig. 1, the model is comprised of two main components: a Variational Autoencoder (VAE) with a Transformer encoder and decoder that turns the data into a unified embedding space, and a Diffusion model that learn the intra- and inter-column patterns of the crash data and synthesize data. With the proposed two-component structure, the model can better handle and learn from the multi-type tabular crash data, and generate synthetic data of outstanding quality.

3.2. Problem definition of crash data synthesis

Each column in the tabular crash dataset is a variable and each row is a record. In this study, the crash dataset contains four types of variables: ordinal, nominal, count, and real-valued. The first three are discrete variables: nominal variables represent categorical data without any inherent order (e.g., the presence of light: 0 = no, 1 = yes); ordinal variables have a natural order (e.g., the hour of the day: 0, 1, 2, ..., 23); and count variables indicate the frequency of occurrences using non-negative integers (e.g., crash frequency: 0, 1, 2, 3, ...). Real-valued variables are continuous variables that can take any value within a range, such as the length of the road segments.

M_{dis} and M_{con} denote the number of discrete and continuous variables, respectively. Each row in the crash data represents a record of crash frequency for a specific road segment at a given hour of the year, along with the corresponding road geometry and environment variables. It is represented as a vector $x = [x^{dis}, x^{con}]$, where x has a dimensionality of $(M_{dis} + M_{con})$. The i th discrete variable x_i^{dis} has C_i candidate values, where $x_i^{dis} \in \{1, \dots, C_i\}$. We input the original training dataset $\mathcal{T} = \{x\}$ to the model to learn the parameterized generative model $p_\theta(\mathcal{T})$, where synthetic data $\hat{x} \in \hat{\mathcal{T}}$ can be generated.

3.3. Crash data transformation using VAE-Transformer

Crash data transformation refers to the process of converting the raw tabular crash data into column embedding. Effective transformation enables the model to extract meaningful information signal from the data (Borisov et al., 2024). The transformation process involves several steps: Feature Tokenization, Transformer Encoding and Decoding, and Detokenization.

Feature Tokenization. Tokenization is the first step to map all the columns into a d-dimensional vector—the initial column embedding. We apply one-hot encoding to preprocess discrete variables, i.e., count, ordinal, and nominal variables, denote as $x_i^{one-hot} \in \mathbb{R}^{1 \times C_i}$. Then each record can be represented as

$$x = [x^{con}, x_1^{one-hot}, \dots, x_{M_{dis}}^{one-hot}] \in \mathbb{R}^{M_{con} + \sum_{i=1}^{M_{dis}} C_i} \quad (1)$$

where x is a $(M_{con} + \sum_{i=1}^{M_{dis}} C_i)$ dimensional vector. Then create a d-dimensional vector for continuous and discrete columns, respectively,

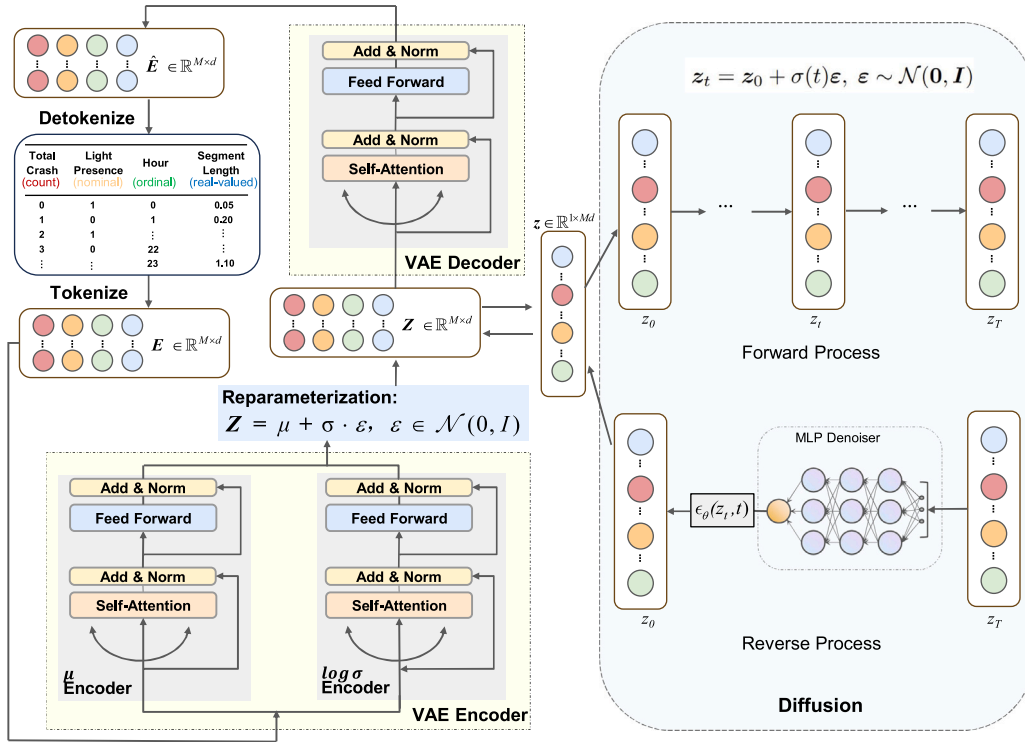


Fig. 1. The hybrid VAE-Diffusion multi-type data generation model architecture for enhancing crash frequency prediction.

by applying linear transformation and creating an embedding lookup table for the two types of columns.

$$e_i^{con} = x_i^{con} \cdot w_i^{con} + b_i^{num}, \quad w_i^{con}, b_i^{num}, e_i^{con} \in \mathbb{R}^{1 \times d} \quad (2)$$

$$e_i^{dis} = x_i^{one-hot} \cdot W_i^{dis} + b_i^{dis}, \quad \begin{cases} b_i^{dis}, e_i^{dis} \in \mathbb{R}^{1 \times d}, \\ W_i^{dis} \in \mathbb{R}^{C_i \times d}. \end{cases} \quad (3)$$

where $w_i^{con}, b_i^{num}, b_i^{dis}, W_i^{dis}$ are learnable parameters of the tokenizer. The initial column embedding E is defined in Eq. (4).

$$E = [e_1^{con}, \dots, e_{M_{con}}^{con}, e_1^{dis}, \dots, e_{M_{dis}}^{dis}] \in \mathbb{R}^{M \times d} \quad (4)$$

Transformer Encoding and Decoding. Simple tokenization may lead to an incomplete understanding of variable characteristics, especially in crash datasets that contain ordinal, nominal, count, and real-valued variables. To address this, we incorporate a Transformer encoder-decoder structure for multi-dimensional encoding, enhancing the model's capability to capture variable dependencies and complex data patterns.

Both the encoder and decoder consist of two-layer transformers, each containing a Self-Attention module and a Feed-Forward Neural Network (FFNN). The FFNN, denoted as H_0 , is implemented as a two-layer Multi-Layer Perceptron (MLP), as described in Eq. (5). The loss function for training the VAE is given in Eq. (6), where $\mathcal{L}_{recon}(x, \hat{x})$ is the reconstruction error between the original input x and its reconstruction $\hat{x}$, $\mathcal{L}_{KL}$ is the Kullback-Leibler (KL) divergence measuring the distance between the learned latent distribution and the standard normal prior distribution, and β is a hyperparameter that balances the trade-off between reconstruction quality and latent space regularization.

$$\begin{cases} H_1 = \text{ReLU}(FC(H_0)) \in \mathbb{R}^{M \times D}, \\ H_2 = FC(H_1) \in \mathbb{R}^{M \times d}. \end{cases} \quad (5)$$

$$\mathcal{L} = \mathcal{L}_{recon}(x, \hat{x}) + \beta \mathcal{L}_{kl} \quad (6)$$

The VAE encoders generate the latent parameters μ and σ , constructing the latent embedding Z according to Eq. (7).

$$Z = \mu + \sigma \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad Z \in \mathbb{R}^{M \times d} \quad (7)$$

Detokenization. Finally, we detokenize and reconstruct each column's token representation back to its original format. The detokenization process mirrors the tokenization step and is defined in Eqs. (8) and (9).

$$\begin{cases} \hat{x}_i^{con} = \hat{e}_i^{con} \cdot \hat{w}_i^{con} + \hat{b}_i^{con}, \\ \hat{x}_i^{one-hot} = \text{Softmax}(\hat{e}_i^{dis} \cdot \hat{W}_i^{dis} + \hat{b}_i^{dis}) \end{cases} \quad (8)$$

$$\hat{x} = [\hat{x}_1^{dis}, \dots, \hat{x}_{M_{dis}}^{dis}, \hat{x}_1^{one-hot}, \dots, \hat{x}_{M_{cat}}^{one-hot}] \quad (9)$$

where $\hat{w}_i^{con} \in \mathbb{R}^{d \times 1}$, $\hat{b}_i^{con} \in \mathbb{R}^{1 \times 1}$, $\hat{W}_i^{dis} \in \mathbb{R}^{d \times C_i}$, and $\hat{b}_i^{dis} \in \mathbb{R}^{1 \times C_i}$ are the learnable parameters of the detokenizer.

3.4. Crash data synthesis using diffusion model

The Diffusion model is designed to learn the underlying distribution of crash data and generate synthetic data. Inspired by the thermodynamic process where particles spread out until evenly distributed, the model adds and spreads out noise to the data. It then learns to reverse this process, removing the noise step by step to create clear, structured samples. Instead of directly learning the distribution of the data itself, the Diffusion model learns how to denoise the data. This strategy allows it to generate diverse data and reduces the likelihood of mode collapse.

After the VAE-Transformer model is well-learned, latent embedding Z can be extracted from the encoder, flattened, and then transformed into a vector as:

$$z = \text{Flatten}(\text{Encoder}(x)) \in \mathbb{R}^{1 \times M d}. \quad (10)$$

The distribution of the latent embeddings $p(z)$ is learned by the diffusion model via two processes: a forward noise-adding process and a reverse denoising process.

In the forward process, the Stochastic Differential Equation (SDE) and Variance Exploding (VE) (Song et al., 2020) are applied to obtain the perturbed data z_t , which is the diffused embedding at time t , and $\sigma(t)$ represents the noise level, as follows:

$$z_t = z_0 + \sigma(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (11)$$

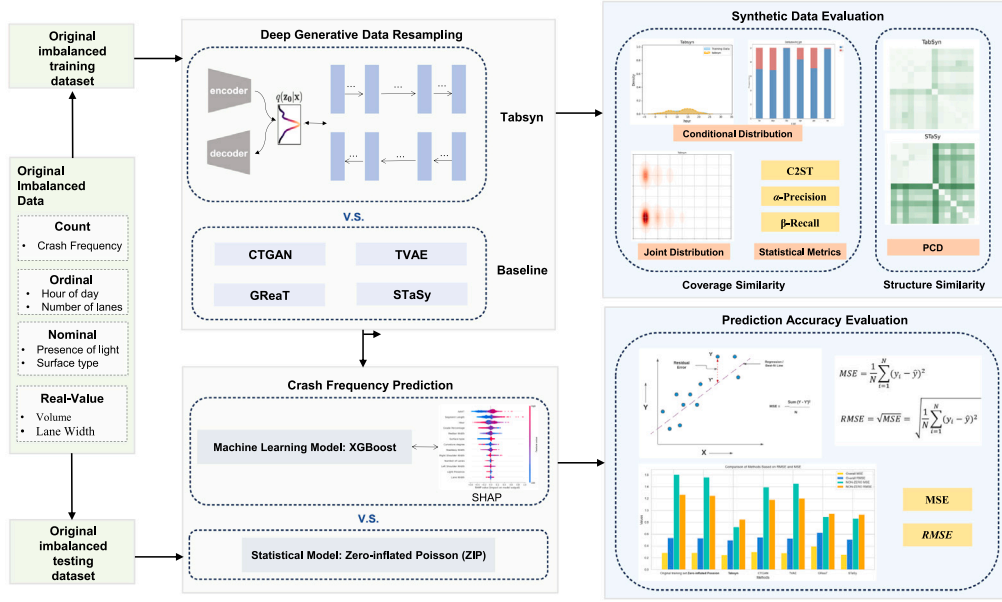


Fig. 2. The framework of data generation and the designed evaluation experiments.

In the reverse process, the score function $\nabla_{z_t} \log p(z_t)$ and the standard Wiener process ω_t are used, as described by:

$$dz_t = -2\sigma(t)\sigma(t)\nabla_{z_t} \log p(z_t)dt + \sqrt{2\sigma(t)\sigma(t)}d\omega_t. \quad (12)$$

The loss function (Karras et al., 2022) is defined as:

$$\mathcal{L} = \mathbb{E}_{z_0 \sim p(z_0)} \mathbb{E}_{t \sim p(t)} \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} \|\epsilon_\theta(z_t, t) - \epsilon\|_2^2, \quad (13)$$

where z_t is obtained from the forward process in (11).

To obtain the denoising function $\epsilon_\theta(z_t, t)$, a five-layer Multilayer Perceptron (MLP) is trained, where each hidden layer is fully connected (Kotelnikov et al., 2022). The architecture is as follows:

$$\begin{cases} h_0 = FC_{in}(z_t) \in \mathbb{R}^{1 \times d_{hidden}}, \\ h_{in} = h_0 \oplus t_{emb}, \\ h_1 = \text{SiLU}(FC_1(h_0)) \in \mathbb{R}^{1 \times 2d_{hidden}}, \\ h_2 = \text{SiLU}(FC_2(h_1)) \in \mathbb{R}^{1 \times 2d_{hidden}}, \\ h_3 = \text{SiLU}(FC_3(h_2)) \in \mathbb{R}^{1 \times d_{hidden}}, \\ h_{out} = FC_{out}(h_3) \in \mathbb{R}^{1 \times d_{in}}. \end{cases} \quad (14)$$

The denoising function is then applied to the loss function in (13) for training.

4. Experimental design

4.1. Resampling and evaluation framework

This study aims to address the excessive zero observations and enhance the overall performance of crash frequency modeling. The data generation and evaluation framework is shown as Fig. 2. We compare the quality of synthetic data generated by both TabSyn and the baseline models. The quality of synthetic data can be assessed based on its coverage and structural similarity to the real data. We adopt various metrics, including distribution plots, Detection Score (C2ST), α -Precision and β -Recall. While the structural similarity is appraised by Pair-wise Correlation Difference (PCD).

We use a machine learning (ML) model, XGBoost, for prediction. To evaluate the modeling accuracy of each resampling method, we compared crash frequency prediction models that were fitted using data balanced by different generative models. The relevant metric, calculated using the test set, is MSE. We also compare the accuracy of the

proposed ML-based model to classic statistical models—Zero-inflated Poisson Regression. Since ML-based models can often be perceived as black boxes due to their complexity, we use SHAP (SHapley Additive exPlanations) to visualize and provide an intuitive explanation of the prediction results.

4.2. Data description

This study focuses on crashes on the mainline of Interstate 5 (I-5) from milepost 139.69 to 210.10 in Washington State in 2019. The crash dataset, sourced from the Highway Safety Information System (HSIS) (Porter et al., 2022), includes roadway and crash files. The roadway file provides information for each road segment, such as the number of lanes, presence of lighting, lane width, segment length, and other details. The crash file contains officer-reported crash incidents.

We select road segments longer than 0.05 miles (approximately 64 m) (Pu et al., 2020) to reduce the potential data coding errors. To learn the impact of time factors on crash frequency, we count the crashes that happened in each segment for every hour of the day throughout the year. The number of crashes on each segment for each hour within a 24-hour period, along with the segment's geometric information, is treated as a record. As a result, there are a total of 3714 crashes across 372 road segments in both directions in 2019. The total number of records is 372 (road segments) * 2 (directions) * 24 (hours) * 1 (year) = 17 856, of which 15 142 (84.8%) have zero observations. The prevalence of zero observations in our dataset indicates a significant excess zero problem. The summary statistics of variables included in this research are shown in Table 1.

4.3. Data resampling

To address excessive zero observations in our dataset, we use the deep generative over-sampling technique (i.e., TabSyn, STaSy, CTGAN, GReaT, and TVAE) to resample our imbalanced crash dataset. We take the training set as the input for the generative models. They then can learn the data distribution and generate synthetic non-zero crash records for rebalancing. The test set is for evaluating the synthetic data quality and prediction accuracy, which should remain the same during the whole modeling process.

Table 2 summarizes the data organization for model training and testing. The original dataset was randomly split into a 70% training

Table 1
Data description.

Variables	Description	Non-Zero records				Zero records			
		Mean	STD	Min	Max	Mean	STD	Min	Max
Dependent variable									
Total_Crash	Crash occurrence per segment per year per hour	1.368	0.796	1	10	0	0	0	0
Explanatory variables									
Ordinal									
Hour	The hour when crashes happened	12.443	5.624	0	23	11.331	7.117	0	23
Number_of_Lanes	Number of lanes (in one direction)	3.552	0.556	2	5	3.484	0.583	2	5
Nominal									
Light_Presence	Presence of light (0 for absence; 1 for presence)	0.299	–	0	1	0.306	–	0	1
Surface_type	Surface type (0 for Asphalt; 1 for Concrete)	0.671	–	0	1	0.595	–	0	1
Real-value									
AAHT	Annual average hourly traffic flow (10 ³ veh/h)	3.570	1.775	0.0001	8.620	2.661	1.751	0.0001	8.6199
Segment_Length	Segment length (mile)	0.188	0.189	0.050	1.100	0.134	0.124	0.050	1.100
Lane_Width	Lane width (ft)	12.467	1.483	11	22	12.355	1.329	11	22
Roadway_Width	Total roadway width for the roadway segment (ft)	87.804	14.760	55	132	85.464	14.443	55	132
Median_Width	Median width of the segment (ft)	56.083	50.650	7	300	60.482	51.853	7	300
Left_Shoulder_Width	Left shoulder width of the segment (ft)	8.530	3.496	0	12	8.970	3.013	0	12
Right_Shoulder_Width	Right shoulder width of the segment (ft)	8.530	3.496	0	12	8.970	3.013	0	12
Curvature_degree	Curvature degree (°)	0.747	0.969	0	4	0.672	0.997	0	4.070
Grade_Percentage	Grade percentage (%)	−0.055	2.250	−5	5	0.123	2.086	−5	5

Table 2
Dataset split and resampling details.

Data type	Original dataset	Test set	Training set	
			Original	Balanced
Zero records	15 142	4543	10 599	10 599
Non-zero records	2714	815	10 599	10 599 (Synthetic: 8700)
Total	17 856	5358	12 498	21 198

set (with 12 498 records) and a 30% testing set (with 5358 records). The training set is balanced using synthetic non-zero crash records. The ratio of zero to non-zero crashes samples is 1:1.

4.4. Model performance evaluation

To validate the effectiveness of the proposed approach, we structured the experiment in two parts. First, we evaluated the quality of the synthetic data generated by different generative models. Second, we compared the prediction accuracy of these data-driven modeling approaches with a statistical model.

4.4.1. Baseline models for comparison

We compare four baseline deep generative models against our proposed multi-type data generation model, including Conditional Tabular Generative Adversarial Networks (CTGAN), tabular variational autoencoder (TVAE), Generation of Realistic Tabular data (GReaT), and Score-based Tabular Data Synthesis (StaSy). These baseline models are chosen as they represent key generative approaches based on Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion models, making them relevant for comparison with our proposed model. Each of these models learns the underlying data distribution differently. CTGAN and VAE are based on two fundamental generative models, namely GANs and VAEs. GReaT, inspired by the transformer-based large language models (LLMs), treats each row of tabular data as a sentence and learns sentence-level distributions. And StaSy is a diffusion-based generative model. We evaluate the quality of the synthetic data they generate. The evaluation focuses on coverage and structural similarity. We then use the data generated by these models to predict crash frequency with XGBoost.

We also compared our data-driven approach with two statistical models: the Zero-inflated Poisson (ZIP) (Lee and Mannering, 2002) and GAM-Poisson (Mao et al., 2021).

The Zero-inflated Poisson (ZIP) model, which is designed to handle count data with excess zeros. It assumes that two separate processes

generate the data: A process that generates only zeros (zero-accident state); and another Poisson process that generates non-negative counts, including zeros. The ZIP model can be defined as

$$P(n_{ij}) = \begin{cases} p_i + (1 - p_i)e^{-\lambda_{ij}}, & n_{ij} = 0, \\ (1 - p_i) \frac{\lambda_{ij}^{n_{ij}} e^{-\lambda_{ij}}}{n_{ij}!}, & n_{ij} = n > 0. \end{cases} \quad (15)$$

where n_{ij} is the number of accidents on the roadway section i during period j ; p_{ij} is the probability that segment i is in the zero-accident state; and λ_{ij} is the expected number of accidents for segment i in period j when it is not in the zero-accident state.

Different from ZIP, the GAM-Poisson framework does not rely on the structural-zero assumption, which makes it more suitable for real-world crash frequency modeling. The GAM-Poisson model assumes that the number of crashes Y_i in each road segment i follows a Poisson distribution (Mao et al., 2021):

$$Y_i \sim \text{Poisson}(\lambda_i \cdot E_i) \quad (16)$$

$$\log(\lambda_i) = \beta_0 + f_1(X_{i1}) + f_2(X_{i2}) + \dots + f_p(X_{ip}), \quad i = 1, \dots, n \quad (17)$$

where E_i is the exposure variable (AAHT), λ_i is the expected crash rate, β_0 is the intercept, X_{ij} represents the j th predictor for observation i , and $f_j(\cdot)$ denotes the smooth function capturing the non-linear relationship between each predictor and crash risk. The smooth functions are implemented using B-spline basis functions with 5 degrees of freedom, which can be expressed as:

$$f_j(X_j) = \sum_{m=1}^{M_j} \beta_{jm} B_m(X_j), \quad j = 1, \dots, p \quad (18)$$

where $B_m(\cdot)$ are the B-spline basis functions and β_{jm} are the estimated coefficients.

4.4.2. Synthetic data quality metrics

The quality of synthetic data is evaluated by how similar its coverage and structure are to the real data. The coverage similarity was evaluated via distribution plots and statistical metrics like Detection Score (C2ST), α -Precision, and β -Recall. Meanwhile, the structural similarity is assessed using the Pair-wise Correlation Difference (PCD).

Classifier Two Sample Test (C2ST) is a metric that evaluates how difficult it is to distinguish real data from synthetic data. To calculate C2ST, first, create a table combining all rows of real and synthetic data, label each row accordingly, split it into training and validation sets, and then train a Logistic Regression machine learning mode to predict these labels. The score is derived from the model's performance, typically

based on the average ROC AUC score across all the cross-validation splits. The C2ST score ranges from 0 to 1, where values closer to 1 indicate greater similarity to the real data, given by

$$\text{score} = 1 - (\max(\text{ROC AUC}, 0.5) \times 2 - 1). \quad (19)$$

α -Precision P_α and β -Recall R_β are two high-order metrics that measure the overall fidelity and diversity of synthetic data (Alaa et al., 2022). Let $\tilde{X}_g$ and $\tilde{X}_r$ represent the synthetic and real samples, and S_r^α and S_g^β denote the α -support of the real distribution and β -support of the synthetic data distribution, respectively. Both metrics range from 0 to 1, where values closer to 1 indicate higher fidelity, diversity, and better coverage, calculated as

$$P_\alpha = \mathbb{P}(\tilde{X}_g \in S_r^\alpha), \quad (20)$$

$$R_\beta = \mathbb{P}(\tilde{X}_r \in S_g^\beta). \quad (21)$$

Pair-wise Correlation Difference (PCD) is another metric used to evaluate structural similarity. Let $S_{A,B}$ and $R_{A,B}$ denote the correlation coefficient between column A and B in the real and synthetic datasets, respectively. The average absolute difference is normalized to ensure it falls within the range of $[0, 1]$, as defined in

$$PCD(S_{A,B}, R_{A,B}) = \frac{|S_{A,B} - R_{A,B}|}{2}. \quad (22)$$

4.4.3. Crash frequency modeling accuracy metrics

Mean Squared Error (MSE) is a commonly used metric to evaluate regression accuracy. MSE represents the average of the squared differences between predicted and observed values, calculated as

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (23)$$

where y_i are the observed values, $\hat{y}_i$ are the predicted values, and n is the number of observations.

4.4.4. Prediction interpretation

SHAP (SHapley Additive exPlanations) provides a unified approach to interpreting machine learning model predictions by assigning each feature an importance value based on its contribution to the prediction. This methodology is grounded in the concept of Shapley values from cooperative game theory, ensuring a fair distribution of the “payout” (model prediction) among the features. The SHAP value of feature i is denoted as ϕ_i , given by

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]. \quad (24)$$

where $S \subseteq N$, where N is the total set of all features, and S is any subset of N that excludes the feature i . The terms $|N|$ and $|S|$ denote the number of features in sets N and S , respectively. The function $f(S)$ represents the model's prediction using only the features in subset S , while $f(S \cup \{i\})$ is the prediction when feature i is added to subset S .

5. Result analysis

5.1. Synthetic data evaluation

The quality of the synthetic data is evaluated by the similarity between the real and synthetic data, including the coverage similarity and the structural similarity. Coverage similarity is how similar are the distributions of two or more variables in both datasets. Structural similarity is how similar are the underlying patterns and correlations between the two datasets (Vallevik et al., 2024). We also use a metric called β -Recall to access the diversity of the synthetic data.

Table 3

Detection Score (C2ST) using Logistic Regression classifier, α -Precision, and β -Recall.

	TabSyn	STaSy	CTGAN	GReaT	TVAE
C2ST	0.9745	0.2327	0.4396	0.7839	0.1428
α -Precision	0.9825	0.8442	0.9648	0.5563	0.4292
β -Recall	0.4838	0.1864	0.2282	0.0000	0.0638

5.1.1. Coverage similarity

Conditional density distribution. Figs. 3–6 compares the distribution of real crash data to synthetic data generated by TabSyn, STaSy, CTGAN, GReaT, and TVAE, corresponding to count, ordinal, real-valued, and nominal variables, respectively. As shown in these figures, deviations in the distribution of synthetic data based on the proposed TabSyn are much smaller than those based on STaSy, CTGAN, GReaT, and TVAE for all variables. For instance, for count variable ‘Total Crashes’ in Fig. 3, the distribution generated by TVAE and GReaT show abnormal concentration at certain values. In Fig. 4, the distribution for the ordinal variable ‘Hour’ generated by STaSy and TVAE, shows a noticeable shift in the peak values compared to the real data. What is more, in Fig. 6 the generated distributions of nominal variables are more accurate with TabSyn, compared to the others.

Joint probability density distribution. Fig. 7 display the joint distribution of between variables Total Crashes and Presence of Light; Total Crashes and Hour; as well as Lane Width and Segment Length, respectively. As shown in Fig. 7(a)–(c), the deviations in the synthetic data distribution generated by TabSyn are much smaller than those based on STaSy, CTGAN, GReaT, and TVAE for all variables. The distribution with TabSyn shows two prominent dense areas in Fig. 7(a) that align well with the real data; while STaSy has significant discrepancies, with only one dense area concentrated in a few narrow bands, indicating it failed to capture the data pattern. Similarly, Fig. 7(b) shows that TabSyn outperforms the other methods, which exhibit various degrees of deviation from the benchmark data, with STaSy and TVAE showing the most significant discrepancies. As for Fig. 7(c), GReaT replicates the main dense area, but with some variations in shape and less concentration; TVAE generated distribution shows several discrete points rather than a continuous distribution; CTGAN generated distribution has some additional noise and less concentration; and STaSy generated distribution has a totally different dense area from the real data.

All in all, synthetic data generated by TabSyn has the most similar distribution across all types of variables — count, ordinal, nominal, and real-valued data — justifying its capability in accurately extract the feature patterns of the data.

In Table 3, we present the results of Detection Score (C2ST), α -Precision, and β -Recall across all methods. For each of these metrics, higher scores indicate better performance. C2ST measures how difficult it is to distinguish real data from synthetic data, with higher scores indicating better performance. TabSyn scores the highest (0.9745) C2ST, indicating it produces synthetic data very similar to real data. α -Precision measures the fraction of synthetic samples that resemble the most typical real samples, where TabSyn also score the highest (0.9825). β -Recall measures the coverage of real samples by synthetic data, and TabSyn scores the highest again (0.4838), demonstrating good coverage. Overall, TabSyn consistently outperforms other methods across all metrics, closely matching real data in terms of fidelity and diversity.

5.1.2. Structural similarity

Pair-wise Correlation Difference (PCD). We plot the PCD heatmap for each pair of columns in our data, as shown in Fig. 8 In the plot, lighter colors indicate smaller PCD values between the real and synthetic data, reflecting more accurate correlation estimations. TabSyn gives the most accurate correlation estimation, while other methods exhibit suboptimal performance.

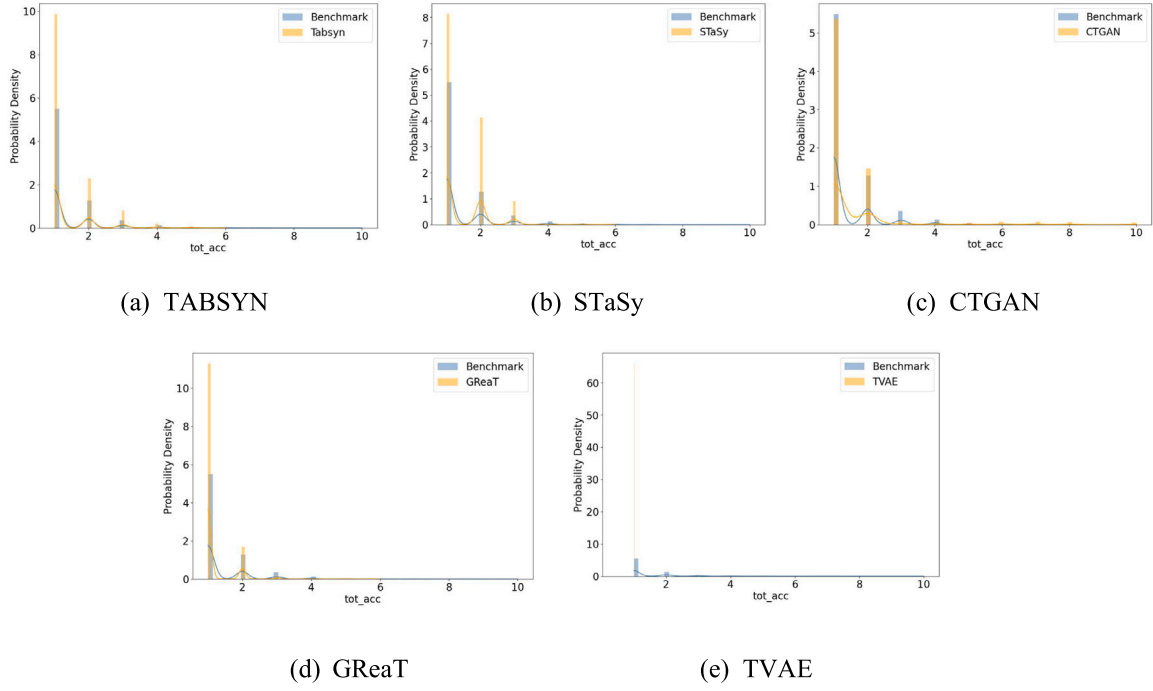


Fig. 3. Comparison of conditional distributions: Total crashes (count variable).

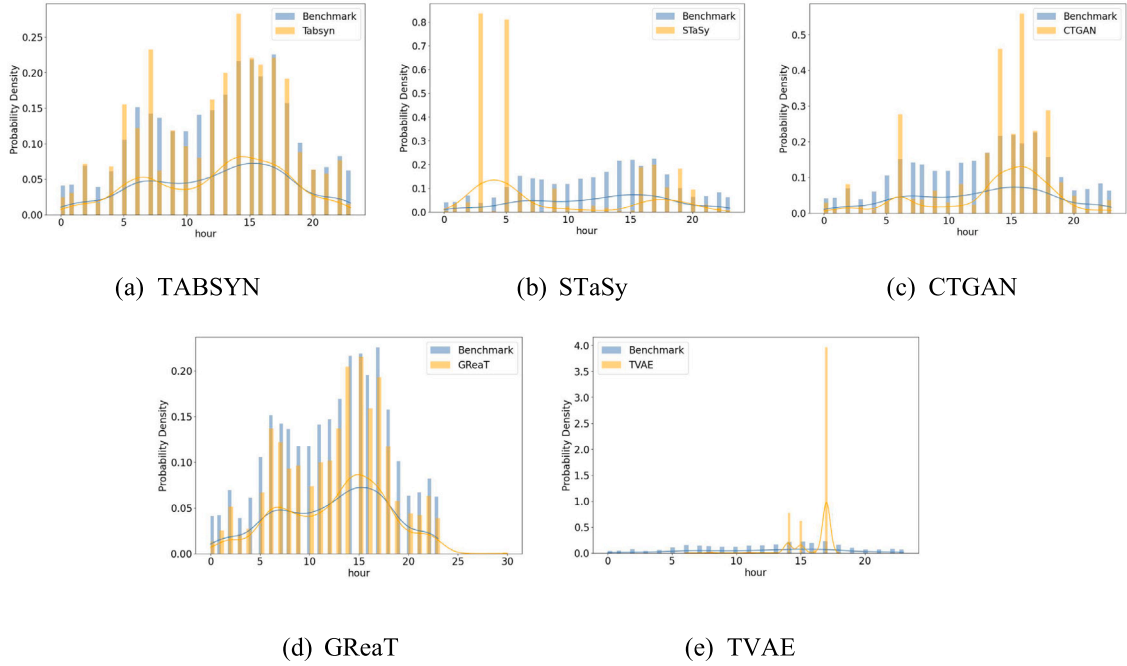


Fig. 4. Comparison of conditional distributions: Hour (ordinal variable).

5.2. Crash frequency modeling prediction accuracy

We employed generative models to oversample the original imbalanced dataset. Subsequently, we compared the accuracy of the fitted crash frequency model using the datasets balanced by different methods, alongside the original imbalanced training set. XGBoost was used as the prediction model, with hyperparameters optimized using grid search with 5-fold cross-validation on the training set.

The results are presented in Table 4. Smaller MSE values indicate better performance. As shown in Table 4, TabSyn demonstrates superior

performance across all evaluation metrics, achieving the lowest MSE at every Top-K threshold.

Among the two statistical models, GAM-Poisson consistently outperforms ZIP across all evaluation metrics. For example, at Top 1%, GAM-Poisson achieves MSE of 23.644, compared to 28.089 for ZIP, representing a 15.8% improvement.

For high-risk segment identification (Top 1%–15%), data-driven machine learning methods outperforms statistical approaches. Among the synthesis-based methods, TabSyn stands out as the best performer, achieving MSE of 14.671 at Top 1%, 7.161 at Top 5%, and 3.823 at Top

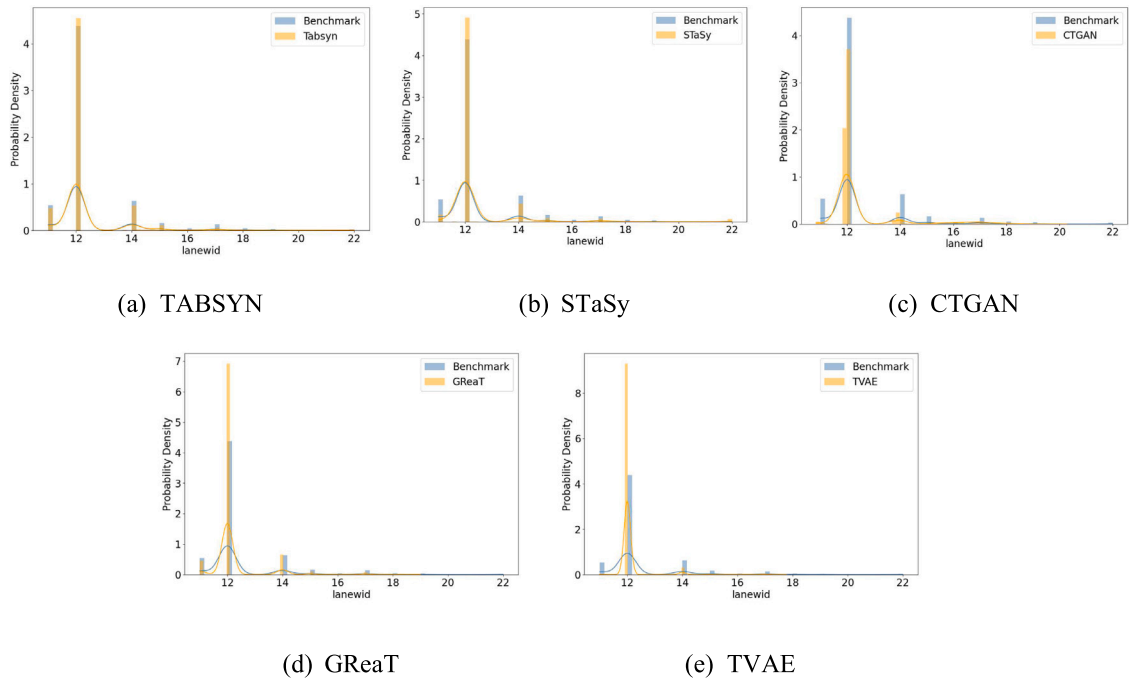


Fig. 5. Comparison of conditional distributions: Lane width (continuous variable).

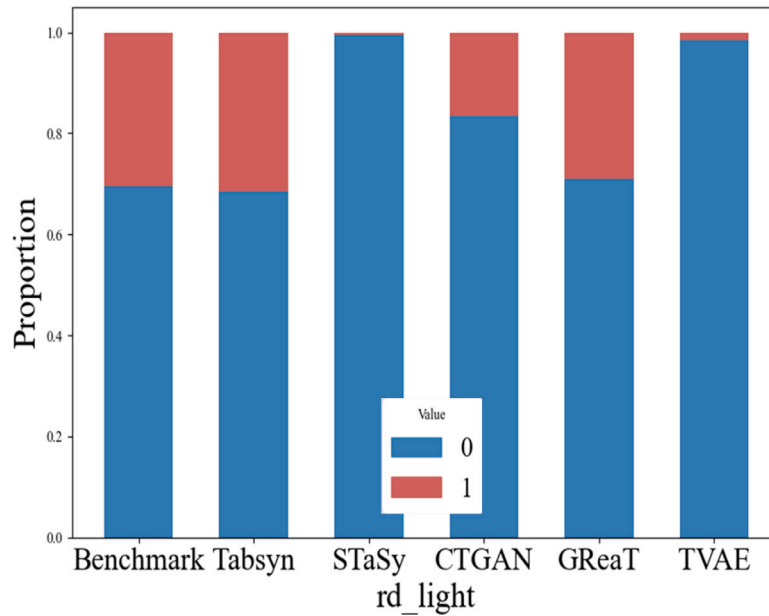


Fig. 6. Comparison of conditional distributions: Presence of light (nominal variable).

15%. Compared to the best statistical model (GAM-Poisson), TabSyn reduces prediction error by 37.9% at Top 1% and 31.3% at Top 5%.

Regarding overall prediction performance, GAM-Poisson demonstrates competitive results with MSE of 0.274. However, TabSyn still slightly outperforms GAM-Poisson with 0.270, achieving the best overall prediction accuracy among all models.

5.3. Discussion on key features

5.3.1. SHAP post-hoc analysis

The SHAP summary plot (Fig. 9) illustrates the impact and direction of various input features on the predicted crash count for a given road segment. The colors represent the magnitude of each feature, with

blue points indicating lower feature values and red points representing higher values. The SHAP values on the x-axis show the features' positive or negative influence on the model's output, where a positive SHAP value suggests the model predicts a higher crash count and a negative SHAP value indicates a lower predicted crash count.

Fig. 9 also ranks the importance of these features from top to bottom, where AAHT is the most influential feature, followed by Segment Length, Hour, Grade Percentage, etc. AAHT (Annual Average Hourly Traffic) has the greatest impact on the predicted crash count, with higher AAHT values (in red) contributing to an increase in crashes. The result is consistent with previous studies (Hou et al., 2021; Høy and Hesjevoll, 2020). This phenomenon is likely explained by the fact that increased traffic volume leads to more vehicle conflicts, while

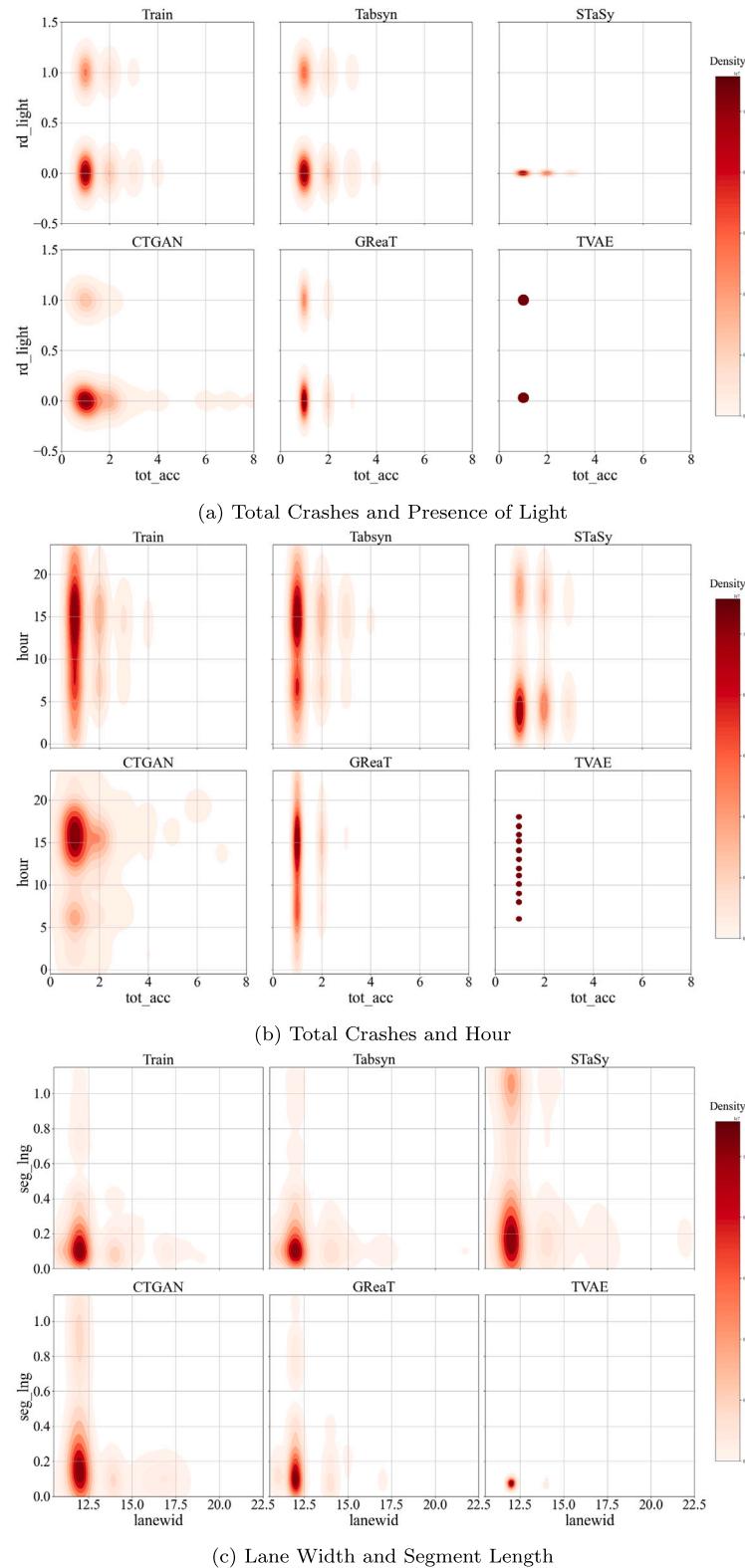


Fig. 7. Joint probability distribution of synthetic data. (a) Total Crashes and Presence of Light, (b) Total Crashes and Hour, (c) Lane Width and Segment Length.

high vehicle density and congestion may reduce available reaction times and increase the likelihood of crashes. Additionally, prolonged congestion or stop-and-go traffic causes driver fatigue, which further elevates crash risk. Segment Length ranks second in importance. Longer road segments (in red) tend to show more crashes simply because

they encompass a larger area, leading to a higher overall crash count due to the greater length being analyzed. The Hour variable is also highly influential. The Hour variable is also highly influential, with rush hour periods (in red) linked to a higher crash count. During these hours, increased traffic flow, time pressure, and driver fatigue raise

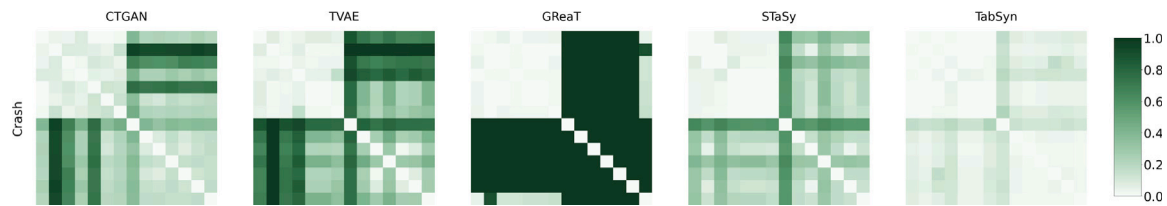


Fig. 8. Pair-wise Correlation Difference of synthetic data and the real data.

Table 4

Prediction accuracy (MSE) on eight crash frequency models.

Model	Top 1%	Top 3%	Top 5%	Top 7%	Top 9%	Top 11%	Top 13%	Top 15%	ALL
Original	26.209	15.129	12.084	10.129	8.566	7.582	6.898	6.374	0.290
ZIP	28.089	16.288	13.120	11.070	9.419	8.388	7.663	7.113	0.310
GAM-Poisson	23.644	13.327	10.671	8.788	7.409	6.592	6.002	5.565	0.274
CTGAN	23.417	13.192	10.524	8.754	7.369	6.546	5.937	5.480	0.274
TVAE	21.722	12.494	9.879	8.121	6.805	6.028	5.453	5.038	0.286
GReaT	19.175	10.484	8.190	6.708	5.552	4.849	4.345	3.993	0.359
StaSy	19.672	11.191	8.664	7.262	6.055	5.344	4.806	4.486	0.366
TabSyn	14.671	8.584	7.161	6.077	5.174	4.866	4.140	3.823	0.270

Note: Best performance values are highlighted in bold.

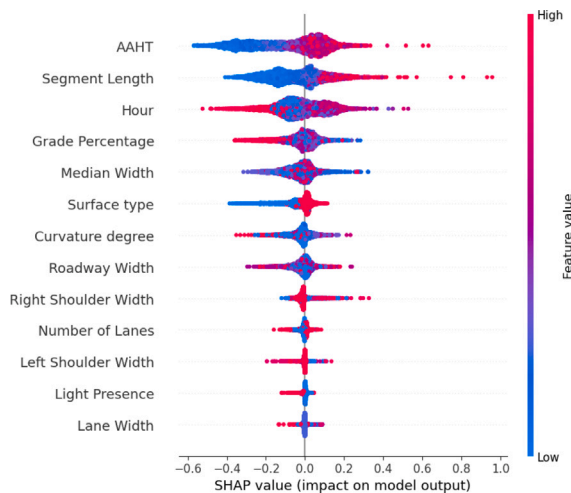


Fig. 9. SHAP value summary plot. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the likelihood of accidents. Additionally, road geometry features like Grade Percentage, median width, surface type, and curvature degree also significantly impact crash count.

Fig. 10 shows the SHAP total effects for Surface Type and Light Presence. The horizontal axis represents the feature value, the left vertical axis shows the SHAP values, and the right vertical axis is color-coded by the joint distribution with AAHT. This plot demonstrates the non-linear effects of features on prediction results. The color gradient represents AAHT levels, ranging from blue (low traffic volume) to red (high traffic volume). This coloring reveals how the SHAP values of one feature (e.g., Surface Type) vary across different traffic volumes, exposing the combined effects of multiple features on crash frequency.

In Fig. 10(a), the x -axis represents road surface type (0: asphalt; 1: concrete). The SHAP values for asphalt roads ($x = 0$) are mostly negative, indicating that asphalt pavement may help reduce the predicted crash count and improve safety. Additionally, for asphalt roads, the negative SHAP effects are more pronounced under high traffic volumes (red dots), with values reaching as low as -0.4 , indicating a positive interaction between high traffic volume and asphalt pavement, i.e., their safety benefits are amplified when combined. Possible explanations

include: higher maintenance frequency on high-traffic roads for maintaining surface smoothness and skid resistance, more cautious driver behavior under high-traffic conditions, and higher design standards for high-traffic asphalt roads such as wider lanes or better drainage, which indirectly improve safety. In contrast, the SHAP values for concrete roads ($x = 1$) are higher, showing that concrete surfaces could increase the predicted crash count. One potential explanation is that worn concrete surfaces can become slick, particularly in adverse weather conditions such as rain or snow, which diminishes skid resistance and elevates the risk of crashes. Furthermore, roads with concrete pavement often undergo less frequent repair and maintenance compared to those with asphalt pavement, which may result in surface deterioration over time and contribute to an increased accident rate.

Fig. 10(b) shows the effect of lighting on the predicted crash count. The x -axis represents light presence (0: light present; 1: no light). Under lit conditions ($x = 0$), the SHAP values can be as low as -0.125 , indicating that road lighting significantly reduces the predicted crash count. Furthermore, under lit conditions, segments with high AAHT exhibit SHAP values clustering in the range of $(-0.125, -0.05)$, indicating that road lighting substantially reduces crash risk on high-traffic roads. Given that high-traffic roads have a higher risk of nighttime crashes, the safety benefits of lighting are more significant on these segments. In contrast, under unlit conditions ($x = 1$), the SHAP values are mostly above -0.05 , indicating higher predicted crash risk when lighting is absent. This confirms that lighting substantially reduces crash risk. Research shows that accidents are more likely in darkness, particularly on unlit roads. Road lighting can reduce nighttime accidents by about 50%, with a slightly greater effect on fatal accidents (Wanvik, 2009).

This analysis suggests several practical safety recommendations. Regular inspections and maintenance should be carried out on concrete roads to prevent deterioration. On road sections with concrete pavement, stricter safety measures, like lower speed limits or barriers, can help reduce crash count. For roads without sufficient lighting, installing more streetlights or reflective markers can improve safety, especially in low-visibility conditions and at night.

The dependence plot for SHAP interaction values helps distinguish main effects from interaction effects. Fig. 11 illustrates the main effects of variables like Hour, AAHT, Lane Width, and Grade Percentage, along with selected interaction effects. Main effects capture the standalone contribution of each feature, while interaction effects reveal how pairs of features together influence the prediction.

Fig. 11(a) shows that the SHAP main effect values between 00:00 and 18:00 are mostly positive when peak crash periods occur between

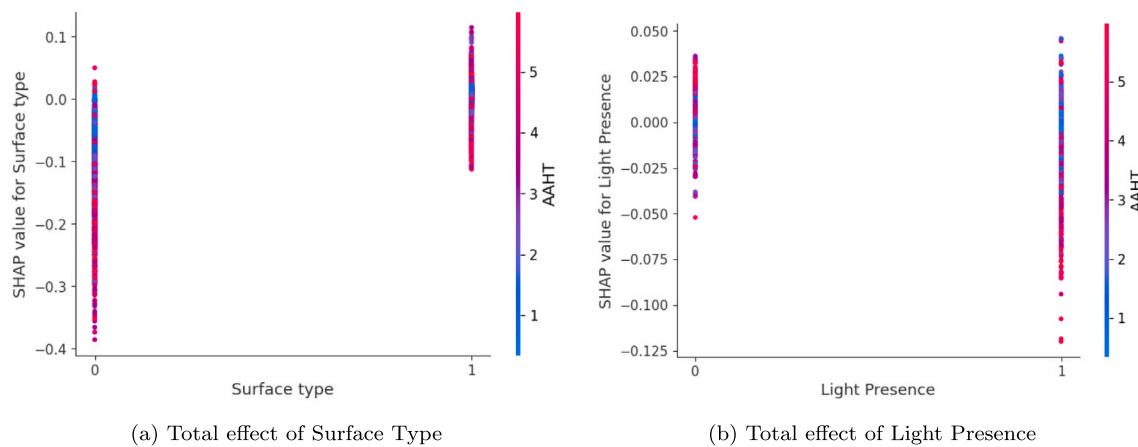


Fig. 10. SHAP total effect plots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

13:00 and 17:00. After 18:00, the values gradually decrease, reaching their lowest point at 21:00, before rising again. Fig. 11(b) indicates that road segments with asphalt pavement significantly reduce predicted crash count during the peak accident period (13:00 to 17:00). One possible explanation for this pattern is the increased traffic volume and driver fatigue during the afternoon peak hours, which contributes to higher crash risk. The reduced crash count on asphalt pavements could be due to better traction and surface conditions, as well as clearer road markings and signage on asphalt surfaces, making it safer compared to concrete during these high-risk periods.

Therefore, safety controls should be enhanced during peak accident hours (13:00–17:00) by increasing traffic monitoring and reducing accident response times. This will help minimize delays and lower the risk of secondary collisions. Asphalt surfaces, which improve vehicle handling and safety, should be prioritized on high-risk road segments to reduce accident rates.

In Fig. 11(c), the main effect of AAHT shows that at low levels of traffic (AAHT between 0 and 2), the accident risk is relatively low. As AAHT increases to a moderate level (2 to 4), the accident risk begins to rise, and at high AAHT levels (above 4), the risk increases sharply. This indicates that higher traffic volumes are associated with a significant increase in predicted crash counts. Fig. 8(d) examines the interaction effect of AAHT and light presence. Under high AAHT conditions, the SHAP interaction values with light presence (in red) are close to or below zero. This suggests that, in high-traffic scenarios, the presence of lighting may reduce accident risk in certain cases. This finding can be attributed to the role of lighting in improving visibility and overall road safety, especially under high traffic conditions. Adequate lighting enhances the visual field, allowing drivers to detect other vehicles more easily. Additionally, lighting helps drivers adapt more quickly to changing light conditions, maintain better alertness, and adhere to traffic rules, which is crucial during periods of high traffic volume. It also aids in minimizing speed-related accidents by providing a clearer view of the road, contributing to overall safer driving behavior in high-traffic scenarios (Wanvik, 2009).

To improve road safety, it may be beneficial to install and optimize street lighting on high-traffic road segments and key intersections. Additionally, in areas with insufficient lighting, using supplementary measures such as reflective signs and warning lights might further reduce accident risks, especially under poor weather or low-visibility conditions.

Fig. 11(e) shows that as Lane Width increases, the main effect value moves closer to zero, indicating that wider lanes reduce accident risk. Since wider lanes offer drivers more space and maneuverability, lowers the chance of vehicle collisions and improves road safety. In Fig. 11(f), narrower lanes exhibit a notable decrease in accident risk when matched with a smaller Curvature Degree. In contrast, wider

lanes mitigate the effects of higher curvature, reducing the potential risk of accidents caused by sharp curves. This suggests that wider lanes offset the impact of curvature on accident risk, helping to reduce the likelihood of crashes.

To improve road safety, increasing lane width on high-risk segments, especially on roads with higher curvature, reduces accident risk. Implementing speed limits and adding clear signage on narrow lanes with low curvature ensures safer driving conditions and decreases the likelihood of accidents.

Fig. 11 shows that when the Grade Percentage is positive (indicating a higher proportion of uphill segments), SHAP values decrease significantly, especially when the Grade Percentage exceeds 3%. This suggests that road sections with more uphill segments experience a lower accident risk, likely due to vehicles moving more slowly as they require more power to ascend. Conversely, when the Grade Percentage is negative (indicating a higher proportion of downhill segments), SHAP values increase, indicating a higher accident risk. This could be because vehicles tend to accelerate more easily on downhill sections, increasing the likelihood of losing control. Fig. 11(h) demonstrates that on road sections with a high proportion of uphill segments (Grade Percentage greater than 3%), wider left shoulders are associated with negative interaction values, further reducing accident risk. The additional space provided by the wider left shoulder likely offers drivers more room for maneuvering or stopping safely in emergencies, thereby lowering the chances of accidents.

On roads with a high proportion of uphill and downhill segments, increasing shoulder width can enhance safety. A wider shoulder provides a larger buffer zone, allowing drivers more room to maneuver, especially during emergencies. This additional space helps to reduce accident risks by offering better control and response options for drivers.

5.3.2. Traffic management recommendation

To enhance road safety, targeted improvements to infrastructure and traffic management strategies should be prioritized. For road infrastructure enhancements, upgrading high-risk road segments with durable materials like asphalt and implementing proactive maintenance schedules for concrete surfaces can improve traction and prevent surface deterioration, particularly in adverse weather conditions. Expanding road lighting coverage at intersections and curves is essential for better nighttime visibility, while widening lanes and shoulders on steep gradients and sharp curves can provide additional maneuvering space to reduce crash risks. In areas where physical expansion is impractical, advanced reflective markers and dynamic signage can serve as effective alternatives. Traffic management measures, such as adaptive monitoring systems and variable speed limits during peak hours, can help alleviate congestion and lower crash risks. Stricter

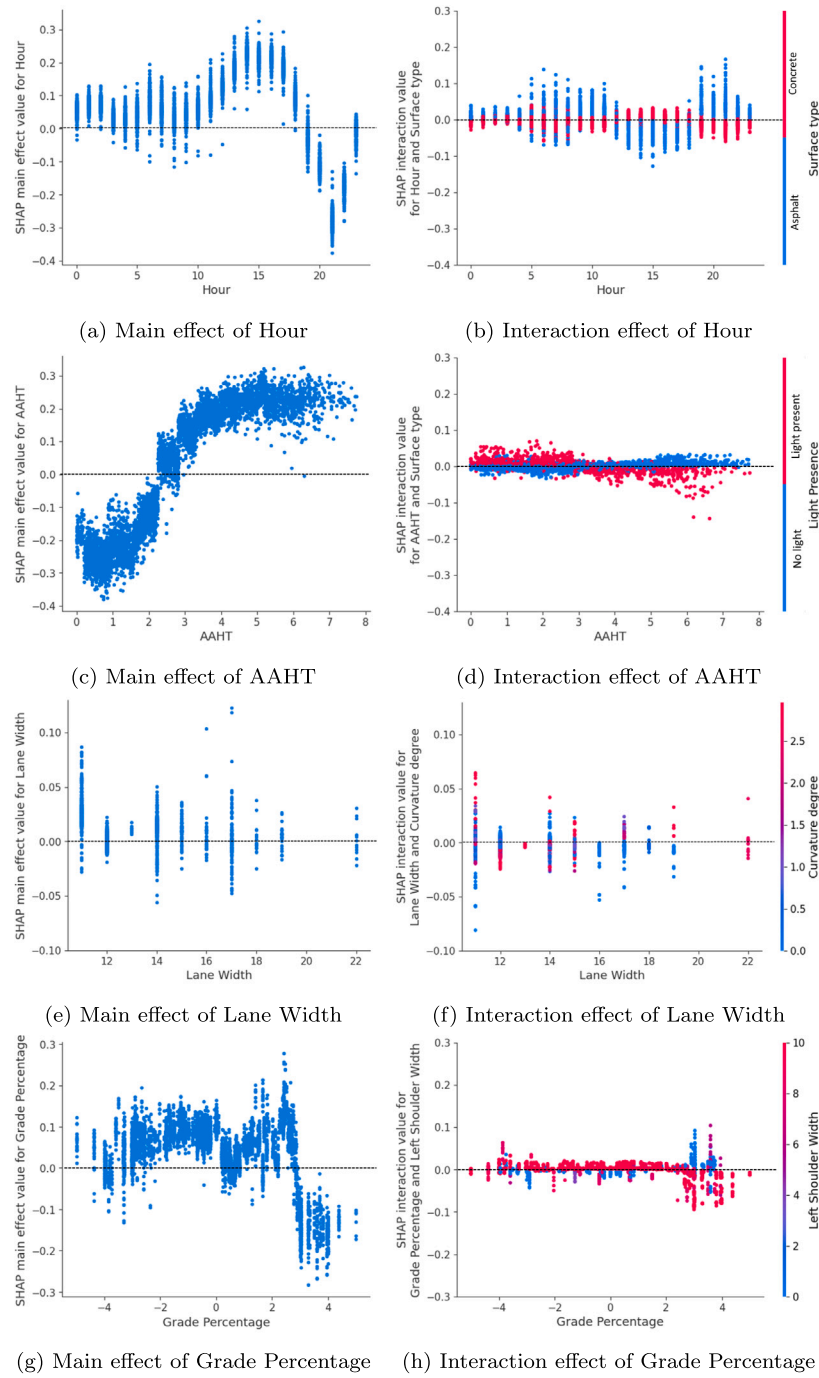


Fig. 11. SHAP main and interaction effects plots. (a) Main effect of Hour, (b) Interaction effect of Hour, (c) Main effect of AAHT, (d) Interaction effect of AAHT, (e) Main effect of Lane Width, (f) Interaction effect of Lane Width, (g) Main effect of Grade Percentage, (h) Interaction effect of Grade Percentage.

safety interventions, including reduced speed limits and protective barriers, should be applied in locations with high crash frequencies. These measures, combined with public education campaigns, aim to create a safer and more resilient transportation network.

6. Conclusions

In this study, to address excessive zero observations and mitigate the limitations of traditional crash data collection methods, we develop a hybrid VAE-Diffusion deep generative model to generate realistic crash data. This method considers the multi-type features of the tabular crash data, i.e. count, ordinal, nominal, and real-valued. Through a

comprehensive evaluation of the quality of synthetic data using various metrics, the proposed TabSyn is verified to have the best generation performance. The synthetic data generated by TabSyn demonstrates the best quality in terms of similarity, accuracy, diversity, and structural consistency. We compare the prediction accuracy with data generated by different deep generative methods, results show that TabSyn has the lowest MSE and RMSE (0.248 and 0.498 for overall predictions; 0.720 and 0.849 for non-zero crash predictions) and surpasses all other methods. In addition, all data-driven models (TabSyn, CTGAN, TVAE, GReaT, and STaSy) show better prediction accuracy compared to the statistical Zero-inflated Poisson (ZIP) model, indicating the proposed XGBoost model provides a higher level of prediction accuracy. We also employ SHAP to interpret crash frequency prediction results, identify

key explanatory factors influencing crash counts, and provide corresponding safety policy recommendations to aid governmental agencies in targeting high-risk areas and optimizing resource allocation.

In future work, we will investigate the effectiveness of the proposed method in balancing crash data that includes omitted variables, heteroscedasticity, and endogeneity issues. What is more, we intend to examine how resampling ratios of the majority zero-crash records to the minority non-zero records impact the prediction accuracy and probability distribution. Furthermore, although this study explored the variables correlation between crash frequency and road factors (e.g., road geometry), traffic volume, and hour, etc. with the proposed VAE-Diffusion deep generative model, future research can delve into the causal effects of these factors on crash frequency.

CRediT authorship contribution statement

Junlan Chen: Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Qijie He:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Pei Liu:** Writing – review & editing. **Wei Ma:** Supervision. **Nan Zheng:** Supervision. **Pan Liu:** Resources, Supervision. **Ziyuan Pu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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